

4. Project Planning Phase

ComicCraft – AI Comic Story Creator

Introduction

The planning phase organizes the implementation work into manageable activities and establishes the sequence required to produce a working ComicCraft application.

Development Sequence

The planned sequence is: create the project structure; configure the Python environment; implement configuration and schemas; implement Gemini integration; implement image generation and mock mode; implement PDF creation; build API routes; build the responsive frontend; connect frontend requests using Fetch API; test the complete workflow; and prepare documentation.

Project Structure Plan

The planned structure contains the ComicCraft root folder, app package, service modules, utility module, static frontend files, generated/images, generated/pdfs, environment configuration files, requirements.txt, run.py, and README.md.

Integration Planning

Gemini story generation is planned before image generation because the panel descriptions and image prompts depend on the generated story. Image files are then passed into PDF generation. The frontend consumes the API results and presents the finished comic.

Testing and Risk Planning

Planning includes checks for invalid Gemini JSON, API errors, rate limits, network failures, invalid inputs, unsafe paths, and environments without a powerful GPU. A development/mock mode is included so the application can run without requiring a powerful GPU.

Execution Plan

The project is intended to run inside a Python virtual environment. The documented commands are to create .venv, activate it, install requirements, copy .env.example to .env, add the Gemini API key, run run.py, and open the local application in a browser.

Project reference: ComicCraft – AI Comic Story Creator